

Technology Stack

Date	29 June 2026
Team ID	XXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack

The chosen technology stack provides a complete environment for data preprocessing, model training, visualization, web application development, and deployment, ensuring efficient implementation of the Smart Agricultural Production Optimization system.

Technology category	Technology Used	Purpose
Programming Language	Python	Core language used for data analysis, machine learning, and backend development
Development Environment	Jupyter Notebook	Model development, experimentation, and testing
IDE	Visual Studio Code / PyCharm	Writing, debugging, and maintaining source code
Data Manipulation	Pandas	Data cleaning, preprocessing, and dataset handling
Numerical Computing	NumPy	Efficient numerical operations and array processing
Data Visualization	Matplotlib	Graphical representation of agricultural data

Data Visualization	Seaborn	Statistical visualization and exploratory data analysis
Machine Learning Library	Scikit-learn	Building, training, and evaluating ML models
Web Framework	Flask	Developing web interface for crop prediction system
Version Control	Git	Tracking changes in source code
Repository Hosting	GitHub	Project collaboration and code management
Cloud Platform	Flask Development Server	Used to run and test the application locally during development
Model Deployment	Pickle (.pkl) + Flask Integration	Saving trained ML model and integrating it into web application for prediction
Dataset	Kaggle Crop Dataset	Historical agricultural data for training and testing
Operating System	Windows 10/11	Environment used for development and execution of the project

Machine Learning Algorithms Used

Algorithm	Purpose
Logistic Regression	Baseline classification model for crop prediction using environmental features
K-Means Clustering	Groups similar soil and climate conditions for pattern analysis
K-Nearest Neighbors	Predicts crop based on similarity with historical agricultural data
Decision Tree	Rule-based and ensemble models for improved accuracy in crop classification